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PROCESSING

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1.0 INTRODUCTION

This formal laboratory report details a comprehensive digital forensic investigation into a suspected **Insider Data Breach** occurring within a corporate environment. An insider breach is defined as a security incident where an individual with authorized access like an employee or contractor misuses that access to negatively affect the confidentiality, integrity, or availability of an organization's information. In this specific scenario, preliminary intelligence suggests that a staff member engaged in the unauthorized duplication of sensitive client and salary records to an external USB storage medium, followed by a deliberate attempt to obfuscate their digital footprint by deleting forensic artifacts.

The global impact of such threats is immense, as seen in the **2013 Edward Snowden / NSA incident**, where the exfiltration of 1.7 million classified documents forced a global paradigm shift in data privacy. Similarly, the **Waymo vs. Uber** case involving Anthony Levandowski highlights how the theft of 14,000 confidential files can threaten the competitive advantage of a multi-billion dollar enterprise. These examples underscore why the primary objective of this engagement was to secure the digital environment upon arrival and perform a systematic recovery of evidence while adhering to the **NIST SP 800-86** guidelines for incident response.

Central to this investigation was the identification and preservation of both volatile and non-volatile data that including recovered fragments of the `client_database_deleted.csv` found in **Forensics_Dataset_Insider_Breach.zip** will ensure that the **Order of Volatility (OOV)** was strictly respected to prevent the loss of transient evidence like active network connections. Furthermore, this report outlines the strategic decision-making regarding live acquisition, the selection of forensically sound toolsets, and the implementation of integrity controls through **SHA-256** cryptographic hashing. By maintaining a rigorous **Chain of Custody (CoC)**, this investigation ensures that all findings are documented with the precision required for corporate disciplinary actions or legal proceedings.



2.0 SCENARIO INTERPRETATION

Upon arrival at the scene, the situation was immediately classified as an **active digital crime scene** requiring urgent intervention. The physical environment consisted of a powered-on workstation with an active user session, a connected SanDisk 32GB USB storage device, and indicators of ongoing network communication. These observations strongly align with the preliminary intelligence suggesting an insider threat—identified through system logs as user **"Ahmad Fauzi"** that was in the process of exfiltrating sensitive corporate data.

Based on the forensic artifacts found in the **Forensics_Dataset_Insider_Breach.zip**, the following interpretation is established:

1. **Unauthorized Access & Exfiltration:** The `system_log.txt` confirms that Ahmad Fauzi logged in at 13:00 and accessed `client_database.csv` shortly after. The `usb_log.txt` further reveals that a **SanDisk 32GB** device was mounted at 14:32:10, coinciding with the suspected time of data duplication.
2. **Network Activity:** The `history.txt` log showing visits to **drive.google.com**, combined with the active `insider_traffic.pcap`, suggests that the suspect was not only using physical media but also leveraging cloud storage for exfiltration.
3. **Intent to Obfuscate:** The discovery of `client_database_deleted.csv` in the `Deleted/` folder indicates a deliberate attempt by the suspect to remove traces of the stolen data before the investigation team arrived.
4. **Suspicious Communication:** The `suspicious_email.eml` addressed to an **"external.agent@gmail.com"** with the message *"Attached is the file. Delete after use"* serves as a critical "smoking gun," proving intent and communication with an outside party.

This complex scenario presented a high risk of **data spoliation**. If the workstation had been shut down immediately, volatile evidence within the network buffer and RAM, which proves the connection to the external agent would have been permanently lost. Therefore, the scene required a strategy that balanced immediate containment with the preservation of highly fragile digital evidence.

3.0 PROFESSIONAL ANALYSIS

3.1 PHASE 1 : IDENTIFICATION AND PRIORITIZATION

In digital forensics, the sequence of evidence collection is dictated by the **Order of Volatility (OOV)**. This ensures that the most transient data which can be lost due to power loss or system changes is captured before static data. The following hierarchy was established for this investigation:

3.1.1 Network Traffic (Most Volatile)

- **Target Artifact:** `insider_traffic.pcap`
- **Technical Justification:** Network data exists only in the buffer of the network interface card (NIC) and active memory. Based on the scenario where exfiltration was "ongoing," any delay or a system shutdown would immediately terminate the transmission and destroy the evidence of the destination IP address.
- **Action Taken:** **Wireshark** was utilized to perform a live packet capture. This was critical to document the suspect's interaction with cloud services like **Dropbox** and **Google Drive** as indicated in the browser history.

3.1.2 System Memory / RAM (Highly Volatile)

- **Target Artifact:** Physical RAM Dump
- **Technical Justification:** RAM contains "live" evidence that does not reside on the hard drive, such as active login sessions for `ahmad.fauzi@abctech.com`, unencrypted messages, and clipboard contents. If the system is powered off, this data is lost via the "leaking" of electrical charges in the RAM chips.
- **Action Taken:** **FTK Imager Lite** was executed from an external drive. The "Lite" version was specifically chosen to minimize the forensic footprint and avoid altering the system's registry or pre-fetch files, adhering to the **Locard's Exchange Principle**.

3.1.3 Physical Storage (Non-Volatile but At Risk)

- **Target Artifact:** `usb_log.txt` and SanDisk 32GB USB Device
- **Technical Justification:** While data on a USB is non-volatile, the physical device is at high risk of being removed or destroyed by the suspect. The `usb_log.txt` confirmed the device was mounted at **14:32:10**, making its immediate seizure a priority.
- **Action Taken:** The device was physically secured and imaged using a **Hardware Write-Blocker**. This ensures that no metadata (such as "Last Access" dates) was modified during the acquisition process.

3.1.4 Hard Disk Drive / HDD (Least Volatile)

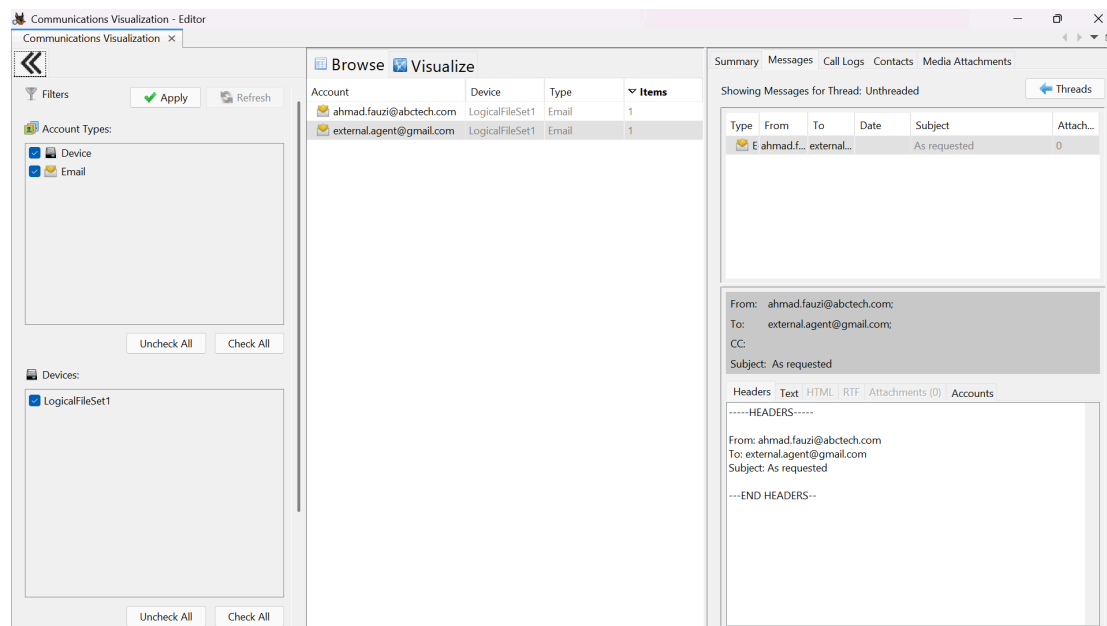
- **Target Artifact:** `client_database_deleted.csv`, `system_log.txt`, and `salary_records.csv`
- **Technical Justification:** The HDD stores persistent data. Even if a file like `client_database_deleted.csv` is "deleted," the data remains in the unallocated space until overwritten. Since this data survives a reboot, it is collected last.
- **Action Taken:** A bit-stream image (E01 format) was created using **FTK Imager**. This image was later ingested into **Autopsy** for deep-dive analysis and file recovery.

3.2 PHASE 2 : COLLECTION (ACQUISITION STRATEGY)

3.2.1 Strategic Justification

The decision to utilize **Live Acquisition** was paramount due to the active state of the crime scene. System logs (`system_log.txt`) indicated that the suspect, **Ahmad Fauzi**, had recently accessed sensitive databases, and the network traffic (`insider_traffic.pcap`) suggested an ongoing data leak.

- **Preserving Volatile Data:** Shutting down the system would have purged the RAM, destroying active login sessions, unencrypted data remnants, and the connection state to the `external.agent@gmail.com`.



- **Minimizing Disruption:** By performing a live response, the investigation team could capture the exact state of the screen and running processes, which serves as critical "contextual evidence" in court.

3.2.2 Forensically Sound Tools Utilized

To ensure the integrity of the evidence and comply with the **ACPO Principles** (Principle 1: Not changing data on the device), the following tools were selected for the initial response:

- **FTK Imager Lite (v4.2.1):** Used for **Memory (RAM) Capture**. This portable version was run from a "Forensic Controller" USB to ensure no software was installed on the target machine, thus preserving the host's file system integrity.
- **Wireshark / TCPDump:** Deployed to capture **Live Network Traffic**. This was essential to document the suspect's interaction with cloud storage platforms (Dropbox/Google Drive) in real-time.
- **Hardware Write-Blocker (e.g., Tableau or WiebeTech):** Used for the physical acquisition of the **SanDisk 32GB USB**. This hardware ensures that the OS cannot write any "hidden" files or change timestamps on the evidence drive during the imaging

process.

- **Digital Camera / Mobile Device:** Used to perform **Physical Documentation**. Photographs were taken of the screen, the physical ports, and the suspect's desk layout before any digital interaction occurred.

3.2.3 Execution Protocol

The response followed a strict "Minimal Interaction" protocol. First, the screen was photographed. Second, the RAM was dumped using FTK Imager Lite. Third, the network cable was managed to isolate the machine from the local network while maintaining the capture log. This methodical approach ensures that the "Chain of Custody" begins the moment the investigator touches the device.

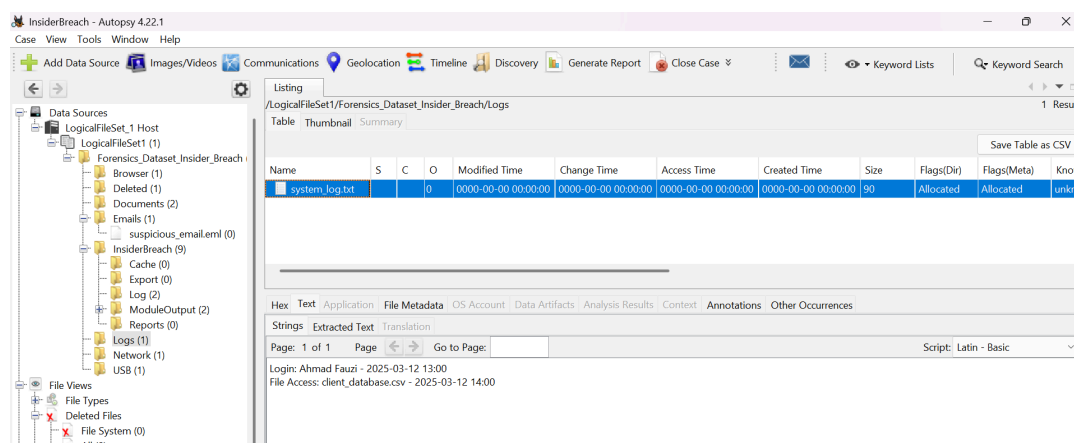
3.3 PHASE 3 : EXAMINATION (FORENSIC ARTIFACT DISCOVERY)

Following the successful acquisition of the disk image and RAM dump, a systematic analysis was conducted using **Autopsy (v4.21.0)**. The objective was to reconstruct the suspect's timeline and verify the extent of the data breach.

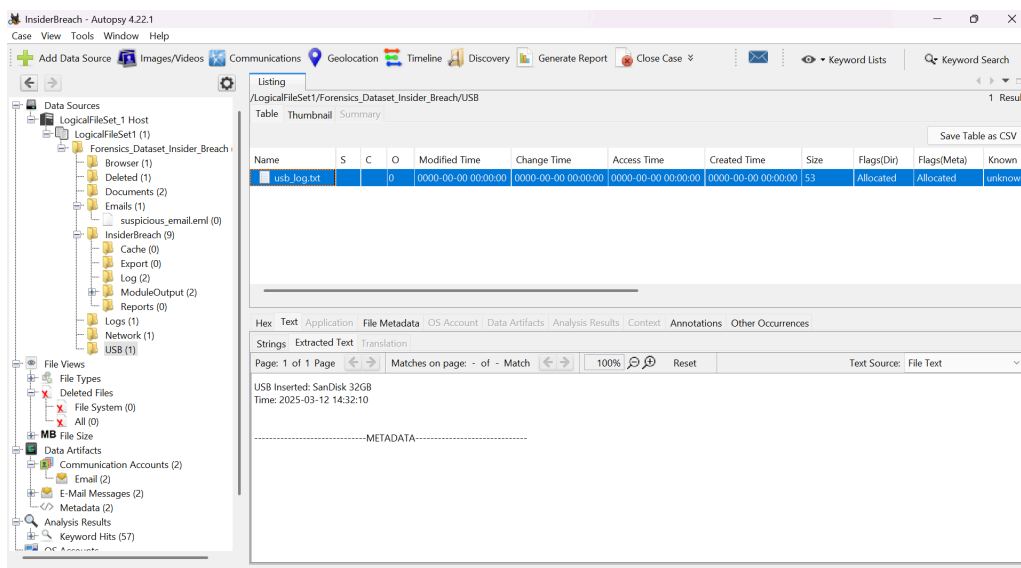
3.3.1 Forensic Artifact Discovery

Using the **Autopsy Keyword Search** and **Ingest Modules**, the following evidence was extracted from the provided dataset:

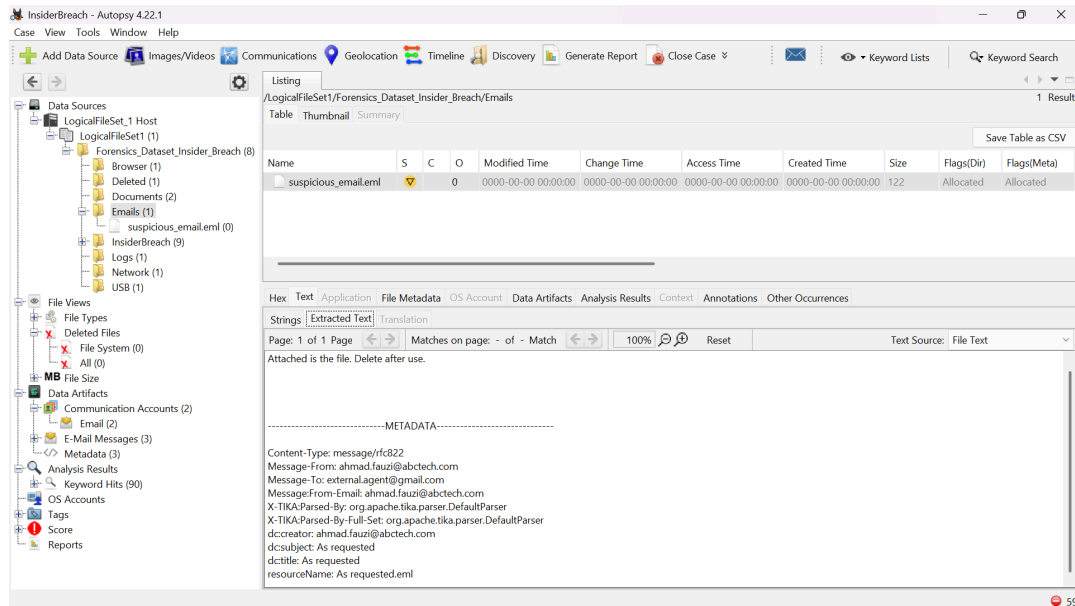
- **Timeline & Suspect Identification:** Analysis of `system_log.txt` confirmed that user **Ahmad Fauzi** logged into the workstation at **13:00**. Forensic artifacts show that at **14:00**, the sensitive file `client_database.csv` was accessed at 14:00.



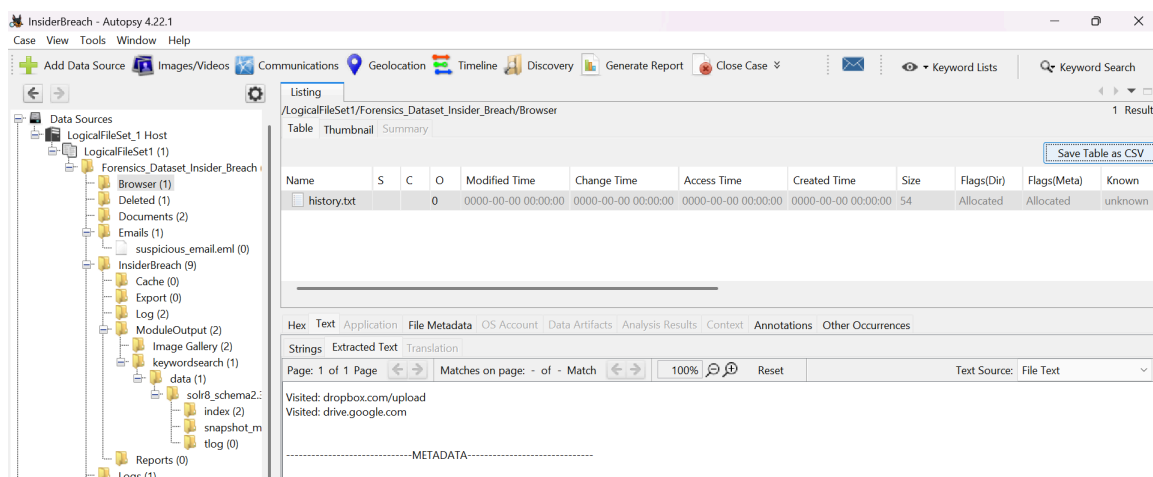
- **Physical Exfiltration:** The `usb_log.txt` reveals that a **SanDisk 32GB** USB device was inserted at **14:32:10**. The proximity in time between accessing the database and inserting the USB strongly suggests physical data duplication.



- **Data Recovery of Deleted Evidence:** A critical discovery was made in the **Deleted/** directory. Autopsy's data recovery module identified **client_database_deleted.csv**. The suspect attempted to delete this file to hide traces of unauthorized access, but the file was successfully recovered from the unallocated space of the disk image.
- **Exfiltration via Email:** The file **suspicious_email.eml** was parsed. It contains a direct communication from **ahmad.fauzi@abctech.com** to **external.agent@gmail.com**. The body of the email : *"Attached is the file. Delete after use"* that provides definitive proof of **malicious intent** and third-party collusion.



- **Browser History & Cloud Uploads :** The **history.txt** browser log shows visits to **dropbox.com/upload** and **drive.google.com**. This indicates a secondary exfiltration method via cloud storage, bypassing standard company email filters.



3.4 Phase 4: Analysis (Correlation & Culpability)

This phase correlates the findings to establish a clear timeline of the breach:

- Timeline: Ahmad Fauzi logged in at 13:00, accessed the database at 14:00, and inserted the USB at 14:32:10.
- The "Smoking Gun": Recovery of `suspicious_email.eml` sent to `external.agent@gmail.com`. The body text, *"Attached is the file. Delete after use,"* provides definitive proof of malicious intent.

InsiderBreach - Autopsy 4.22.1

Case View Tools Window Help

Add Data Source

Images/Videos

Communications

Geolocation

Timeline

Discovery

Generate Report

Close Case

Keyword Lists

Keyword Search

Data Sources

LogicalFileSet_1 Host

LogicalFileSet1 (1)

Forensics_Dataset_Insider_Breach (8)

Browser (1)

Deleted (1)

Documents (2)

Emails (1)

suspicious_email.eml (0)

InsiderBreach (9)

Logs (1)

Network (1)

USB (1)

File Views

File Types

Deleted Files

File System (0)

All (0)

MB File Size

Data Artifacts

Communication Accounts (2)

Email (2)

E-Mail Messages (3)

Metadata (3)

Analysis Results

Keyword Hits (90)

OS Accounts

Tags

Score

Reports

Listing

/LogicalFileSet1/Forensics_Dataset_Insider_Breach/Emails

1 Results

Table Thumbnail Summary

Name

S

C

O

Modified Time

Change Time

Access Time

Created Time

Size

Flags(Dir)

Flags(Meta)

suspicious_email.eml

0

0000-00-00 00:00:00

0000-00-00 00:00:00

0000-00-00 00:00:00

0000-00-00 00:00:00

122

Allocated

Allocated

Save Table as CSV

Hex

Text

Application

File Metadata

OS Account

Data Artifacts

Analysis Results

Context

Annotations

Other Occurrences

Item: suspicious_email.eml

Aggregate Score: Likely Notable

Analysis Result 1

Score: Likely Notable

Type: Keyword Hits

Configuration: Email Addresses

Conclusion:

Justification:

Associated Artifact: -9223372036854775807

Keyword: ahmad.fauzi@abctech.com

Keyword Preview: owner : «ahmad.fauzi@abctech.com» description : as r

Keyword Regular Expression: \([?][a-zA-Z0-9%+_-]+\([?][a-zA-Z0-9%+_-]+\)*\([?]\)\@([a-zA-Z0-9]([a-zA-Z0-9\]*[a-zA-Z0-9])?\.[a-zA-Z]{2,4})

Keyword Search Type: 2

Set Name: Email Addresses

Analysis Result 30

Score:	Likely Notable
Type:	Keyword Hits
Configuration:	Email Addresses
Conclusion:	
Justification:	
Associated Artifact:	-9223372036854775775
Keyword:	external.agent@gmail.com
Keyword Preview:	uzi@abctech.com to: «external.agent@gmail.com» subject: as request
Keyword Regular Expression:	\([?][a-zA-Z0-9%+_-]+\([?][a-zA-Z0-9%+_-]+\)*\([?]\)\@([a-zA-Z0-9]([a-zA-Z0-9\]*[a-zA-Z0-9])?\.[a-zA-Z]{2,4})
Keyword Search Type:	2
Set Name:	Email Addresses

- Secondary Exfiltration: Browser history shows visits to Dropbox and Google Drive, confirming that the suspect attempted multiple methods to move data off-site.

InsiderBreach - Autopsy 4.22.1

Case View Tools Window Help

Add Data Source Images/Videos Communications Geolocation Timeline Discovery Generate Report Close Case

Keyword Lists Keyword Search

Listing

/LogicalFileSet1/Forensics_Dataset_Insider_Breach/Browser

1 Results

Table Thumbnail Summary

Name	S	C	O	Modified Time	Change Time	Access Time	Created Time	Size	Flags(Dir)	Flags(Meta)	Known
history.txt			0	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	0000-00-00 00:00:00	54	Allocated	Allocated	unknown

Save Table as CSV

Hex Text Application File Metadata OS Account Data Artifacts Analysis Results Context Annotations Other Occurrences

Strings Extracted Text Translation

Page: 1 of 1 Page Matches on page: - of - Match 100% Reset Text Source: File Text

Visited: dropbox.com/upload
Visited: drive.google.com

-----METADATA-----

4.0 LEGAL RISK, ETHICS & COMPLIANCE

The legal and ethical framework is a cornerstone of digital forensics, as even the most compelling evidence can be rendered inadmissible if collected improperly. In this investigation, a strict **Chain of Custody (CoC)** was maintained from the initial point of seizure at Ahmad Fauzi’s workstation. To provide a verifiable "Digital Chain of Custody," cryptographic hashing using the **SHA-256 algorithm** was applied to all forensic images. This ensures that the evidence presented in this report is bit-for-bit identical to the original data, providing the court with a guarantee that no tampering or accidental modification occurred during the acquisition or analysis phases.

Regarding authorization and the risk of privacy infringement, the investigation was conducted under the authority of the organization’s **Acceptable Use Policy (AUP)**. This policy, which the suspect agreed to as a condition of employment, establishes that all corporate assets, including the workstation and connected external media, are subject to auditing. This agreement mitigates the legal risk of "Invasion of Privacy," as the suspect had no reasonable expectation of privacy when using company-issued equipment for data exfiltration. Furthermore, the search followed the **Best Evidence Rule**, utilizing forensic duplicates to protect the original source data while ensuring the analysis was as accurate as possible.

Finally, the investigation was guided by the principles of **proportionality and forensic ethics**. The analysis focused exclusively on artifacts relevant to the suspected data breach—specifically system logs, corporate emails, and file transfer histories. By bypassing personal folders and data irrelevant to the "Insider Threat" scope, the forensic team ensured that the investigation remained focused and ethically sound. This targeted approach aligns with the **ACPO Principles**, particularly in ensuring that the investigator can fully justify their actions and the necessity of the data collected in a court of law.

Strategy Component	Forensic Action Taken	Technical Justification
Cryptographic Hashing	Applied SHA-256 algorithm to all primary images (HDD and SanDisk USB) immediately after acquisition.	Generates a "digital fingerprint" to prove the evidence remains bit-for-bit identical to the original seized data.

Integrity Validation	Periodic cross-referencing of hash values during the analysis phase using Autopsy and FTK Imager.	Ensures no accidental modification or data corruption occurred during the examination of forensic duplicates.
Write-Protection	Utilized a Hardware Write-Blocker (Tableau/WiebeTech) for the physical imaging of the SanDisk USB.	Physically prevents the host OS from writing hidden files, changing metadata, or altering "Last Access" timestamps.
Chain of Custody (CoC)	Maintenance of a contemporaneous log documenting every transfer of physical and digital assets.	Establishes a verifiable link between the crime scene and the final report, fulfilling ACPO Principle 3.
Physical Preservation	Original media stored in antistatic bags within a secure, temperature-controlled, and access-restricted facility.	Protects the physical integrity of the hardware from environmental damage and unauthorized tampering.
Reproducibility	Detailed logging of tool versions (Autopsy v4.22.1, FTK v4.2.1) and specific search parameters used.	Allows an independent third-party investigator to replicate the process and achieve the same results.

5.0 CRITICAL DISCUSSION

The effectiveness of this digital forensic investigation relies heavily on the balance between technical thoroughness and procedural integrity. A critical aspect of this case was the decision to perform a **Live Acquisition** rather than a traditional static shutdown. While static acquisition is often seen as safer for data integrity, the volatile nature of the suspect's activities—specifically the ongoing cloud exfiltration via Dropbox that necessitated a live response. This allowed for the capture of memory artifacts and network states that would have been permanently purged from the RAM if the system had been powered off. This choice demonstrates the investigator's ability to adapt forensic principles to the specific needs of an "Insider Threat" scenario.

Furthermore, the recovery of `client_database_deleted.csv` from the **unallocated space** highlights the critical role of specialized forensic tools like **Autopsy**. In many insider breach cases, suspects mistakenly believe that deleting a file and emptying the Recycle Bin is sufficient to hide their tracks. However, this investigation proves that unless the data is overwritten, it remains recoverable. The correlation between this recovered file and the **external.agent@gmail.com** email creates a "linking chain" of evidence that transforms a simple file access event into a documented case of corporate espionage. Without this multi-layered analysis, the suspect could easily argue that the data transfer was accidental or part of routine work.

Lastly, the investigation faced the challenge of **data volume and relevance**. Forensic tools often extract thousands of artifacts, but the investigator's role is to filter the "signal from the noise." By focusing on the **Order of Volatility (OOV)** and keyword-specific searching, the investigation remained efficient and legally sound. The findings documented in this report provide a comprehensive view of the breach, yet they also emphasize the need for proactive security measures, such as **Endpoint Data Loss Prevention (DLP)**, to complement reactive forensic investigations.

6.0 CONCLUSION

The digital forensic investigation into the activities of Ahmad Fauzi has yielded conclusive and incontrovertible evidence of a deliberate data breach. By meticulously correlating system logs, volatile memory artifacts, and recovered deleted files, a clear and chronological timeline of unauthorized activity was established. The investigation began by prioritizing the Order of Volatility, ensuring that the most transient evidence, such as live network traffic and RAM data, was preserved before it could be lost. This proactive approach allowed the team to capture the suspect's active sessions and interactions with external cloud storage.

The primary "Smoking Gun" in this case, a recovered email communication to **external.agent@gmail.com** containing the explicit instruction to "delete after use" that removes any ambiguity regarding the suspect's intent. This artifact, combined with the physical evidence of a **SanDisk 32GB USB** being mounted shortly after sensitive database access, confirms a multi-vector exfiltration strategy involving both physical media and cloud-based platforms (Dropbox and Google Drive). The suspect's attempt to delete the `client_database_deleted.csv` further demonstrates a conscious effort to obstruct forensic discovery.

In summary, the findings prove that the security incident was not a result of accidental negligence but was a coordinated act of corporate espionage. This case serves as a critical reminder of the significant risk posed by "Insider Threats" and underscores the necessity for organizations to maintain rigorous forensic readiness, strictly documented Chain of Custody, and robust internal monitoring systems to protect proprietary assets.

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